Supplementary Material for “Drift Detection, Diagnosis and Reliability Analysis for Deployed Machine Learning Models”

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Section, figure, table and equation numbers without the prefix S refer to the main paper.

This supplement gives four pieces of detail that support the main paper: how the size of the delayed audit affects the diagnosis, the remediation triage in every real setting, the mapping of the ACS relationship codes across the 2019 redesign and the dashboard through which the pipeline reports its results.

S1 Size of the delayed audit

The audit stage of Section 3.3 separates concept drift from label noise with a few hundred late labels. The paper uses 600 labels throughout. Fig. S1 shows how the share of correctly diagnosed windows changes with the number of audit labels on ACS Income design A with logistic regression (LR).

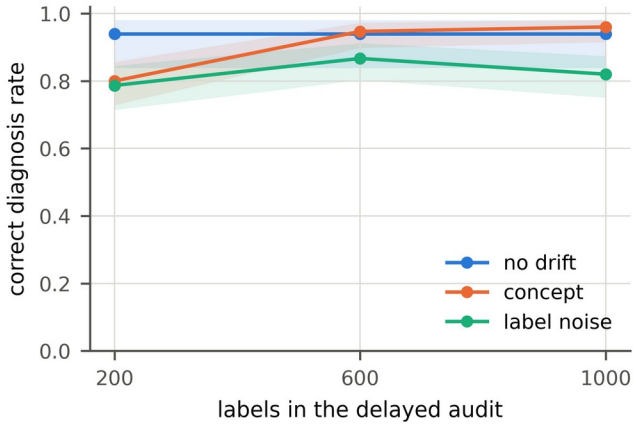


Fig. S1 Share of correctly diagnosed windows on ACS Income design A with LR against the number of audit labels. The shaded bands are 95% Wilson intervals

Concept drift is identified in 80% of windows with 200 labels and 95% with 600, while label noise is identified in 79% and 87%. With 1,000 labels neither rate rises much further (96% and 82%), so the 600 labels used in the paper lie where the gain from more labels has largely flattened. The control windows are diagnosed correctly in the same share of cases at every audit size.

S2 Remediation triage by setting

Section 5.5 summarizes how often the triage of Section 3.5 gives the right advice. Table S1 shows the result for every real setting and both model families.

Table S1 Remediation triage on the real windows. n is the number of windows; Significant gains counts the windows in which a strategy significantly raised holdout accuracy, by importance weighting (IW), retraining on the most recent labeled window (Recent) or retraining on labels of the window itself (Labels); n/a marks settings without an earlier window. Escalated counts the windows in which the triage recommended a retrain, Truly helps those in which a cheap retrain did help and Correct the windows in which the advice was right

			Significant gains			Triage		
Model	Settings	n	IW	Recent	Labels	Escalated	Truly helps	Correct
LR	ACS Income, temporal	8	4/8	7/7	5/8	7	7	8/8
	ACS Income, states	4	0/4	n/a	4/4	0	0	4/4
	ACS Employment, temporal	8	4/8	7/7	2/8	7	7	8/8
	ACS Employment, states	4	2/4	n/a	1/4	1	2	3/4
	CLEAR-10	9	0/9	4/8	6/9	4	4	7/9
	Gas sensor	8	1/8	7/7	8/8	7	7	8/8
GB	ACS Income, temporal	8	0/8	7/7	4/8	7	7	8/8
	ACS Income, states	4	0/4	n/a	3/4	0	0	4/4
	ACS Employment, temporal	8	0/8	7/7	0/8	6	7	7/8
	ACS Employment, states	4	0/4	n/a	0/4	0	0	4/4
	CLEAR-10	9	0/9	8/8	9/9	8	8	9/9
	Gas sensor	8	1/8	5/7	8/8	7	6	7/8

The pattern behind the summary in the paper is visible here. Wherever an earlier labeled window exists, retraining on it helps in almost every temporal window and the triage escalates accordingly. The geographic windows of the census tasks have no earlier window. Importance weighting rarely helps there. The triage advises against retraining in most of them. The remaining wrong verdicts fall mostly on CLEAR-10 with LR and on the geographic ACS Employment windows.

S3 Relationship codes across the 2019 redesign

From 2019 the Census Bureau replaced the relationship variable RELP with RELSHIPP, which splits spouses and partners by sex [1]. To compare windows before and after 2019, the RELSHIPP codes are mapped back to RELP by matching the code labels of the 2018 and 2019 data dictionaries. Table S2 lists the mapping that the code applies.

Table S2 Mapping from RELSHIPP (2019 on) back to RELP (2014 to 2018)

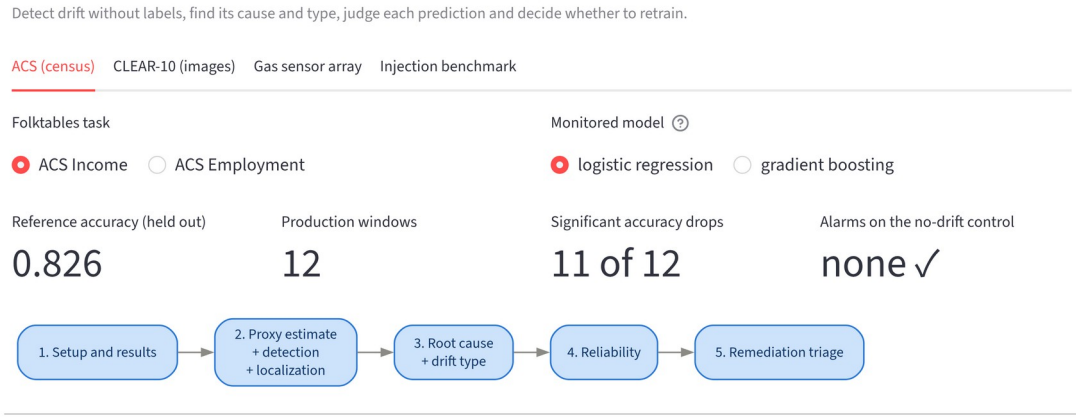
RELP	Label (2018)	RELSHIPP codes (2019 on)
0	Reference person	20
1	Husband/wife	21, 23
2	Biological son or daughter	25
3	Adopted son or daughter	26
4	Stepson or stepdaughter	27
5	Brother or sister	28
6	Father or mother	29
7	Grandchild	30
8	Parent-in-law	31
9	Son-in-law or daughter-in-law	32
10	Other relative	33
11	Roomer or boarder	none
12	Housemate or roommate	34
13	Unmarried partner	22, 24
14	Foster child	35
15	Other nonrelative	36
16	Institutionalized group quarters population	37
17	Noninstitutionalized group quarters population	38

Same-sex and opposite-sex spouses both map to husband/wife and both partner codes to unmarried partner. RELP 11 (roomer or boarder) has no successor, so it never occurs from 2019 onward. This removed category is itself a documented change that the localization can detect.

S4 The monitoring dashboard

Every stage of the pipeline writes its results to one file, which an interactive dashboard presents to the monitoring team. Fig. S2 shows two of its views for ACS Income with LR.

(a)



(b)

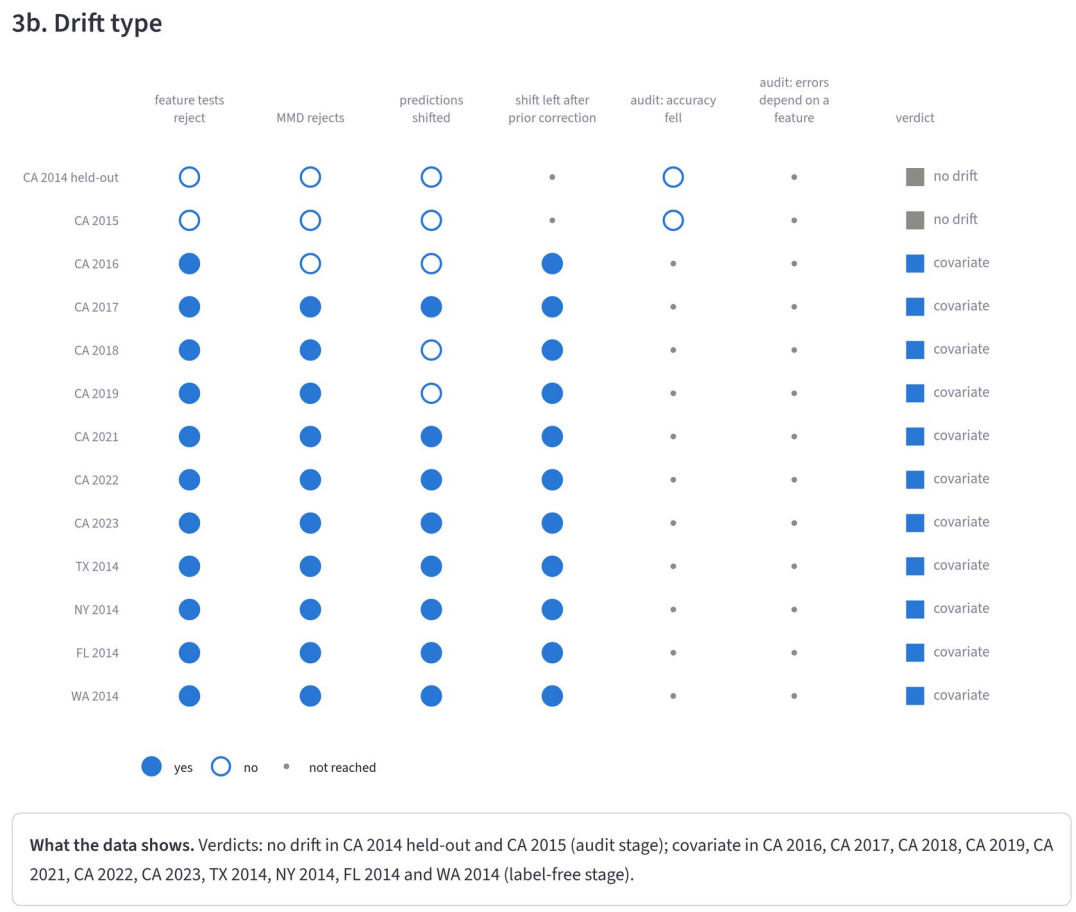


Fig. S2 The dashboard of the pipeline for ACS Income with LR. (a) The overview: dataset, task and model selectors, the headline results and the five steps of the pipeline, each of which opens a detailed view. (b) The drift type step: for every window, the answer to each question of the decision path in Fig. 2 of the paper (filled circle yes, open circle no, small dot not reached) and the resulting verdict, followed by a summary sentence computed from the results

Panel (b) is the decision path of Fig. 2 applied to real windows. CA 2015 passes every label-free test. The audit then finds no significant accuracy drop, so the verdict is no drift. In CA 2016, CA 2018 and CA 2019 the predicted labels do not move, yet the feature tests reject and the shift survives the prior correction, so these windows are covariate shift. The remaining windows answer yes to every label-free question and are also covariate shift. Each view of the dashboard opens with the question it answers and ends with such a summary sentence, so a team can read the diagnosis without reading the code.

References

[1] U.S. Census Bureau "American Community Survey 2019 PUMS Data Dictionary", [online] Available at: https://www2.census.gov/programs-surveys/acs/tech_docs/pums/data_dict/PUMS_Data_Dictionary_2019.csv [Accessed: 21 October 2026]